

Pehan Amarathunge

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EDUCATION

University of Oxford, St John's College

October 2026 - September 2030

DPhil in Applied Mathematics

- Supervised by Prof. Ian Griffiths and Prof. Mohit Dalwadi.
- Funded via a Clarendon Scholarship.

University of Cambridge, Girton College

October 2025 - June 2026

Master of Advanced Study (MASt) in Applied Mathematics (Part III)

Distinction

- **Relevant Modules:** Perturbation Methods, Fluid Dynamics of Climate, Non-Newtonian Fluids, Biological Fluids, Fluid Dynamics of Solid Earth, Slow Viscous Flows.
- **Essay** - *The Scallop Theorem*, supervised by Prof. Eric Lauga (92%).

University College London

September 2022 - June 2025

Bachelor of Science in Mathematics and Mathematical Physics

First Class Honors

- **Weighted Average** - 92.34 %
- **Relevant Modules:** Real Fluids (90.76 %), Geophysical Fluids (99.80 %), Asymptotic Methods

RESEARCH EXPERIENCE

Viscoelastic fluids in Medium Amplitude Oscillatory Shear (MAOS) (UCL)

Summer 2025

- Undertook a 8 week summer project **supervised by Prof. Helen Wilson** and funded by a UKRI Departmental Summer Studentship.
- Investigated the theoretical underpinnings of a model equation by Dr. Peter Laity for the stress response of certain viscoelastic fluids under MAOS experiments.
- Reviewed a novel framework by Randy Ewoldt for characterizing non-linear rheological behavior under an imposed MAOS strain.
- Investigated various constitutive relations in an attempt to match the empirical observations.

Lightning Solvers in Sloshing Modes (UCL)

Summers 2024/2025

- Undertook a 10 week summer project **supervised by Prof. Ted Johnson** and funded by the London Mathematical Society Undergraduate Research Bursary scheme.
- Generalized the Lightning Laplace solver (developed by Nick Trefethen) to deal with eigenvalue boundary conditions by solving a generalised eigenvalue problem instead of a least squares approach.
- Utilized this method to determine the sloshing modes and frequencies of a two-dimensional impermeable tank of arbitrary shape.
- Obtained up to 10 digits of accuracy, when comparing numerical results with existing analytical solutions for specific geometries.

AWARDS AND SCHOLARSHIPS

Clarendon Scholarship | *University of Oxford*

October 2026

Scholarship to fund a DPhil at the University of Oxford.

M.T. Meyer Taught Masters Achievement Scholarship | *University of Cambridge, Girton College*

June 2026

An award of £400 for achieving a Distinction on Part III of the Mathematical Tripos.

Ellen Watson Memorial Scholarship | *UCL*

October 2025

Awarded to the best final year student in Applied Mathematics

Awarded to the top 5% of graduating students in the faculty

UKRI Summer Vacation Studentship | *UKRI - UCL*

July 2025

Awarded a bursary of £2480 to fund a summer research project.

London Mathematical Society - Undergraduate Research Bursary | *UCL*

July 2024

Selected as one of 2 undergraduates in the UCL mathematics department and one of 32 mathematics undergraduates in the UK to receive an award of £2200 to undertake a research internship in the summer.

Stevenson Prize | *UCL*

September 2023

Awarded to the best first year student in applied mathematics out of a cohort of over 300 students

TALKS AND CONFERENCES

1. Part III Seminar Series | Cambridge

December 2025

Title: Viscoelastic Instabilities: Instability without Inertia

2. British Conference of Undergraduate Research (BCUR) | Newcastle

April 2024

Title: Lightning Solvers in Sloshing Modes

3. Tomorrow's Mathematicians Today (TMT) | Online

March 2024

Title: D'Alembert's Paradox and the Drag Crisis

PROGRAMMING SKILLS

Proficient: Python, Matlab, $\text{\LaTeX}$

Python Libraries: NumPy, SciPy, Matplotlib, Pandas, Sympy

PROJECTS

Free Surface Flows Generated by a Moving Pressure Distribution (UCL)

July 2024 - August 2024

- Collaborated with a team of 4 undergraduates to investigate wave patterns generated by a moving pressure distribution, supervised by Dr. Sergei Timoshin.
- Incorporated both gravity and capillary effects and presented our findings in a departmental colloquium.

Addressing the Hare Eats Lynx Paradox (University of Colombo)

June 2023 - August 2023

- Undertook a 8 week summer project supervised by Dr. Kushani de Silva to review and generalise existing predator-prey models for the populations of Canadian Lynx and Snowshoe Hare, by including effects such as: fear, group defense, protection zones, hunting cooperation and epidemics.
- Numerically simulated and compared these models using python and MATLAB to other models from other research papers and communicated my findings in weekly research meetings.

Investigating the Stability of the Three Body Problem (UCL)

May 2023 - June 2023

- Liaised with a team of 4 undergraduates to compare the sensitivity of a three-body system to linear perturbations in various initial conditions using Python.
- Presented our findings in a departmental symposium, implementing the Manim python library for animations and visualisations.

TEACHING EXPERIENCE

Intercollegiate Teaching (University of Oxford)

- Michaelmas Term 2026: Teaching Assistant for Part B Viscous Flows

Transition Mentor (UCL, Department of Mathematics)

September 2024 - December 2024

- Responsible for weekly meetings to help my tutees progress with problem sheets, extend their learning from lectures and prepare them for university examinations and the transition to university life
- Nominated as a finalist for "Transition Mentor of the Year" for my efforts.